



Girraween High School

2020

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 2

Total Marks: 100

Section 1 (Pages 2 – 4) **10 Marks**

- Attempt Q1 - Q10
- Allow about 15 minutes for this section

General Instructions

- Reading time: 5 minutes
- Working time: 3 Hours
- Write using a black or blue pen
- Board approved calculators may be used
- Laminated reference sheets are provided
- Answer multiple choice questions by completely colouring in the appropriate circle on your multiple choice answer sheet on the front page of your answer booklet.
- In questions 11-16 start all questions on a separate page in your answer booklet and show all relevant mathematical reasoning and/or calculations.

Section 2 (Pages 5-11) **90 marks**

- Attempt Q11 - Q16
- Allow about 2 hours and 45 minutes for this section

Section 1 (10 marks)

Attempt Questions 1-10

Allow about 15 minutes for this section

Question 1

$$\int x^3 \cos x \, dx =$$

- (A) $-x^3 \sin x + 3 \int x^2 \sin x \, dx$ (B) $-x^3 \sin x - 3 \int x^2 \sin x \, dx$
(C) $x^3 \sin x - 3 \int x^2 \sin x \, dx$ (D) $x^3 \sin x + 3 \int x^2 \sin x \, dx$

Question 2

$$\int \frac{e^x}{\sqrt{1 - e^{2x}}} \, dx =$$

- (A) $\frac{1}{2} \ln(1 - e^{2x}) + C$ (B) $\sin^{-1}(e^x) + C$ (C) $\tan^{-1}(e^x) + C$
(D) $\cos^{-1}(e^x) + C$

Question 3

$$2e^{\frac{5\pi i}{6}} =$$

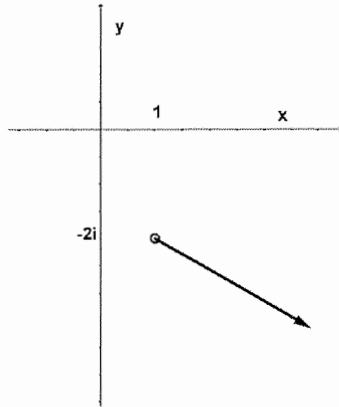
- (A) $\sqrt{3} - i$ (B) $\sqrt{3} + i$ (C) $-\sqrt{3} - i$ (D) $-\sqrt{3} + i$

Multiple choice continues on the following page

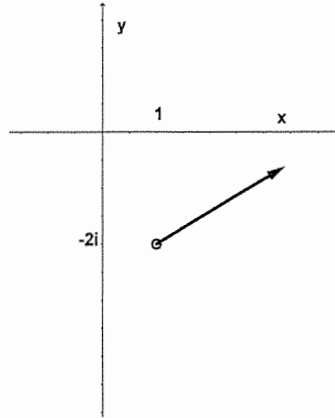
Question 4

Which of the following diagrams shows $\text{Arg}(z - 1 + 2i) = -\frac{\pi}{3}$?

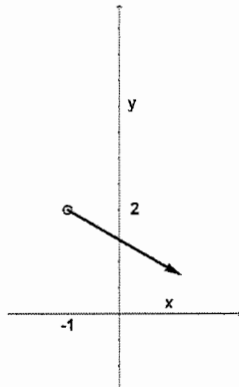
(A)



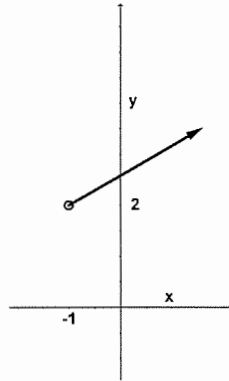
(B)



(C)



(D)



EA

Question 5

The contrapositive of “If it barks it’s a dog” is

(A) “If it doesn’t bark it isn’t a dog”

(B) “If it’s a dog it will bark”

(C) “If it doesn’t bark it’s a dog”

(D) “If it isn’t a dog it doesn’t bark”

Question 6

Which of the following is true?

(A) $\forall a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+ : b = a^3$

(B) $\forall a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+ : b = \sqrt[3]{a}$

(C) $\forall a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+ : a = b^3$

(D) $\forall a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+ : b = a^3$

Multiple choice continues on the following page

Question 7

If a and b are *entirely imaginary* then which of the following is true

- (A) $a^2 + b^2 \geq 2ab$ (B) $a^2 + b^2 \leq 2ab$ (C) $a^2 + b^2 = 2ab$
(D) Any of the above can happen.

Question 8

A particle moves in a straight line. At one point, $x = 6$, $v = 8$ and $a = 18$. An equation of motion for the particle could be

- (A) $v^2 = \frac{x^3}{3} - 8$ (B) $v^2 = x^2 + 28$ (C) $v = x + 2$ (D) $v^2 = 3x^2$

Question 9

A particle moves with simple harmonic motion so that $v^2 = 27 - 18x - 9x^2$. The period and amplitude of the motion are

- (A) Period = $\frac{\pi\sqrt{2}}{3}$ seconds, amplitude = $3m$ (B) Period = $\frac{2\pi}{3}$ seconds, amplitude = $2m$
(C) Period = $\frac{3\pi}{2}$ seconds, amplitude = $3m$ (D) Period = $\frac{\pi\sqrt{2}}{3}$ seconds, amplitude = $2m$

Question 10

The cartesian equation of the line $\underline{i} + 2\underline{j} - \underline{k} + \lambda(2\underline{i} + 3\underline{j} + 4\underline{k})$

is

- (A) $\frac{x+1}{2} = \frac{y+2}{3} = \frac{z+1}{4}$ (B) $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{4}$ (C) $x - 2 = \frac{y-2}{3} = z + 4$
(D) $x + 2 = \frac{y+2}{3} = z + 4$

Examination continues on the following page

Section II (90 marks)

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Start the answers to each question on a separate page in your answer booklet.

In Questions 11-16 your responses should include all relevant mathematical reasoning and/ or calculations.

Question 11 (15 marks)

Marks

(a) If $z = 1 - i\sqrt{3}$

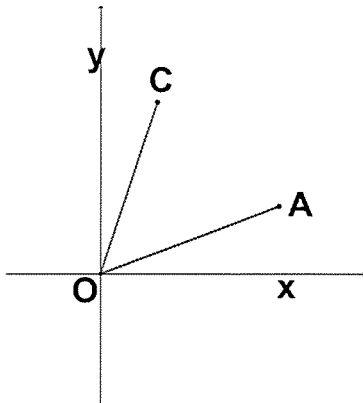
(i) Express z in modulus/argument form

2

(ii) Find z^3 and show that it is real.

2

(b) If O is the origin, $\overrightarrow{OA} = 2 + i$ and $\overrightarrow{OC} = 1 + 2i$ (see diagram)



(i) Find B so that $OABC$ is a rhombus.

1

(ii) By finding $\frac{\overrightarrow{OB}}{\overrightarrow{OA}}$, show that $\tan \angle AOB = \frac{1}{3}$

2

(iii) HENCE show that $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{4}$

3

Question 11 continues on the following page

Question 11 (continued)**Marks**

(c) (i) By letting $(x + iy)^2 = 3 - 4i$, find $\sqrt{3 - 4i}$

3

(ii) Hence solve the equation $z^2 + (4 - i)z + (3 - i) = 0$

2**Question 12 (15 marks)****Marks**

(a) (i) Express $\frac{-13x-10}{(x+1)^2(x-2)}$ in the form $\frac{A}{(x+1)^2} + \frac{B}{(x+1)} + \frac{C}{(x-2)}$

3

(ii) Hence find $\int \frac{-13x-10}{(x+1)^2(x-2)} \cdot dx$

1

(b) Find $\int \frac{1}{\sin x - \cos x - 1} \cdot dx$

3

(c) Find $\int e^x \cos x \cdot dx$

2

(d) (i) Show that $\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$

1

Let $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x \cdot dx$

(ii) Show that $I_n = \frac{1}{n-1} - I_{n-2}$

2

(iii) Hence find I_6

2

(iv) Given that $I_n \rightarrow 0$ as $n \rightarrow \infty$, find $\lim_{n \rightarrow \infty} 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \dots$

1

Examination continues on the following page

Question 13 (15 marks)**Marks**

(a) Prove *by contraposition* that if $n^2 + 6n$ is even, then n is even. 3

(b) Prove *by contradiction* that $\log_3 11$ is irrational. 3

(c) Prove by induction that $3^n \geq n^2$ for all positive integers $n \geq 1$ 3

(d) Prove for all integers x, y that if $10x + y$ is divisible by 17, $3y - 4x$ is also divisible by 17. 2

(e) (i) Prove $a^2 + b^2 \geq 2ab$ for all $a, b \in \mathbb{R}$ 1

(ii) Hence or otherwise prove $a^4 + b^4 + c^4 + d^4 \geq 4abcd$ for all $a, b, c, d \in \mathbb{R}$ 2

(iii) Hence or otherwise prove $\frac{w+x+y+z}{4} \geq \sqrt[4]{wxyz}$ for all $w, x, y, z > 0$. 1

Question 14 (15 marks)

(a) If $\underline{p} = \underline{i} + 2\underline{j} - \underline{k}$ and $\underline{q} = 2\underline{i} - \underline{j} + \underline{k}$

(i) Find $\underline{p} \cdot \underline{q}$ 1

(ii) Find the angle between $\underline{p}$ and $\underline{q}$. 1

(iii) Find $\text{Proj}_{\underline{p}} \underline{q}$ 1

Question 14 continues on the following page

Question 14 (continued)**Marks**

(b) (i) Show that the point $\begin{pmatrix} 12 \\ -6 \\ 6 \end{pmatrix}$ lies on the sphere

1

$$(x - 10)^2 + (y + 12)^2 + (z - 14)^2 = 104.$$

(ii) Show that the line $\begin{pmatrix} 12 \\ -6 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix}$ forms a diameter of the sphere

2

$(x - 10)^2 + (y + 12)^2 + (z - 14)^2 = 104$ and find the other point at which the line intersects with the sphere.

(iii) Show that the line $\begin{pmatrix} -8 \\ 12 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -8 \\ 5 \end{pmatrix}$ is skew to the line $\begin{pmatrix} 12 \\ -6 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix}$

4

(iv) Find the points of intersection of $\begin{pmatrix} -8 \\ 12 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -8 \\ 5 \end{pmatrix}$ and the sphere

3

$$(x - 10)^2 + (y + 12)^2 + (z - 14)^2 = 104.$$

(v) Show that the line $\begin{pmatrix} -8 \\ 12 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -8 \\ 5 \end{pmatrix}$ passes directly above the centre of

2

the sphere $(x - 10)^2 + (y + 12)^2 + (z - 14)^2 = 104$ and find the point at which this happens.

Examination continues on the following page

Question 15 (15 marks)**Marks**

(a) The depth of the water at a wharf is regulated by the tide and can be modelled using simple harmonic motion. If at low tide at 7:00a.m. the depth is $5m$ and at high tide at 1:30p.m. the depth is $13m$

(i) Letting the time be measured in hours and $t = 0$ hours to be 7:00a.m. write a rule for the depth (x) in terms of time (t). **2**

(ii) A certain boat can only reach the wharf when the depth is *greater* than $8m$. What are the times this can happen between 7:00a.m. and the next low tide? **2**

(b) A $5kg$ projectile is launched at a speed of $600m/s$ at an angle of 30° up from the horizontal. It experiences gravity of 50 Newtons and air resistance opposite to its direction of motion of $\frac{5}{6}v$ Newtons.

(i) Show that $\ddot{x} = -\frac{1}{6}\dot{x}$ and $\ddot{y} = -10 - \frac{1}{6}\dot{y}$ where x is horizontal displacement and y is vertical displacement. **1**

(ii) Show that the initial velocities in the horizontal and vertical directions are $300\sqrt{3}m/s$ and $300m/s$ respectively. **1**

(iii) Show that $\dot{x} = 300\sqrt{3}e^{-\frac{t}{6}}$ and $x = 1800\sqrt{3}(1 - e^{-\frac{t}{6}})$ **3**

(iv) Find the maximum possible horizontal range of the projectile (the range it can never quite reach). **1**

(v) Show that $\dot{y} = 360e^{-\frac{t}{6}} - 60$ and $y = -2160e^{-\frac{t}{6}} - 60t + 2160$. **3**

(vi) Find the maximum height of the projectile. **2**

Examination continues on the following page

Question 16 (15 marks)**Marks**

(a) A projectile is launched vertically upwards from the ground at a speed of Um/s . It experiences acceleration due to gravity of $g m/s^2$ and acceleration due to air resistance of $kv^2 m/s^2$.

(i) If x is the vertical height of the projectile above the ground, show that **3**

$$x = \frac{1}{2k} \ln \left(\frac{g+kU^2}{g+kv^2} \right).$$

(ii) Show that the maximum height reached is $\frac{1}{2k} \ln \left(1 + \frac{kU^2}{g} \right)$ metres. **1**

(iii) The projectile starts to fall from its maximum height. It continues to experience acceleration due to gravity of $g m/s^2$ and air resistance *against* its motion of $kv^2 m/s^2$. Letting down be positive, and the point where the projectile reaches its maximum height be $x = 0$, find the *terminal velocity* of the projectile in terms of k and g and show that $x = \frac{1}{2k} \ln \left(\frac{g}{g-kv^2} \right)$. **4**

(iv) Letting T be the terminal velocity, W be the impact velocity (the speed at which the projectile hits the ground) and keeping U as the initial launch velocity, show that $\frac{1}{U^2} + \frac{1}{T^2} = \frac{1}{W^2}$. **2**

Question 16 continues on the following page

Question 16 (continued)**Marks**

(b) (i) Solve $z^5 - 1 = 0$.

1

(ii) If w is the root of $z^5 - 1 = 0$ with the smallest positive argument, show that $w^2 + \frac{1}{w^2} + w + \frac{1}{w} = -1$.

2

(iii) Hence show that $x = \cos \frac{2\pi}{5}$ is a root of the equation $4x^2 + 2x - 1 = 0$.

1

(iv) Hence find the exact value of $\cos \frac{2\pi}{5}$.

1

END OF EXAMINATION!!!

p11

Solutions: Y12 Trial Exam Ext 2 2020 p.1

New Syllabus.

Multiple Choice:

Q. (1) C (2) B (3) D (4) A (5) D (6) A (7) B (8) A (9) B (10) B

$$(1) \int x^3 \cos x \cdot dx \quad \begin{array}{l} u = x^3 \quad v = \sin x \\ u' = 3x^2 \quad v' = \cos x \end{array}$$

$$= x^3 \sin x - 3 \int x^2 \sin x \cdot dx \quad (C)$$

$$(2) \int \frac{e^x}{\sqrt{1-e^{2x}}} \cdot dx \quad (B)$$

$$= \sin^{-1}(e^x) + C$$

$$[\text{By } \int \frac{f'(x)}{\sqrt{1-(f(x))^2}} \cdot dx = \sin^{-1}(f(x)) + C]$$

$$(3) 2e^{\frac{5\pi i}{6}} \quad (D)$$

$$= 2 \times \left[\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right]$$

[by Euler's theorem]

$$= 2 \left[-\frac{\sqrt{3}}{2} + \frac{i}{2} \right]$$

$$= -\sqrt{3} + i$$

$$(4) (A)$$

(5) "If it isn't a dog it doesn't bark" (D)

(6) (A) "For all positive integers a, there is a positive integer b such that $b = a^3$ "

[Note: (B) & (C) said that the cube root of every positive integer was also a positive integer & (D) didn't say a was an integer.]

$$(7) (B) \text{ If } a = xi, b = yi$$

 x, y real then

$$a^2 + b^2 = (xi)^2 + (yi)^2 = -(x^2 + y^2)$$

$$\& 2ab = -2xy$$

As $x^2 + y^2 \geq 2xy$, x, y real

$$-(x^2 + y^2) \leq -2xy$$

$$(8) \text{ Using } a = \frac{d}{dx} \left[\frac{1}{2} v^2 \right] \quad (A)$$

$$\text{in (A), } v^2 = \frac{x^3}{2} - 8$$

$$\frac{1}{2} v^2 = \frac{x^3}{4} - 4$$

$$a = \frac{d}{dx} \left[\frac{1}{2} v^2 \right] = \frac{x^2}{2} \ln \text{ this case, } x=6, a = \frac{6^2}{2} = 18 \&$$

$$v^2 = \frac{6^3}{2} - 8 \Rightarrow v = 8$$

ln (B) $a = x$ & would = 6, not 18ln (C) $a = x+2$ & would = 8, not 18

$$\ln (D) v^2 \neq 3x^2 [8^2 \neq 3 \times 6^2]$$

$$(9) v^2 = 27 - 18x - 9x^2$$

Amplitude: $v^2 = 0$

$$27 - 18x - 9x^2 = 0$$

$$x^2 + 2x - 3 = 0$$

$$(x+3)(x-1) = 0 \quad x = -3 \text{ or } 1 \quad (B)$$

Amplitude = 2.

$$\text{Period: } v^2 = 27 - 18x - 9x^2$$

$$a = \frac{d}{dx} \left[\frac{1}{2} v^2 \right] = -9(x+1)$$

$$\frac{d}{dx} \text{ By } x = -n^2 x, \text{ Period} = \frac{2\pi}{n^2}$$

$$(10) \text{ As } x = 2\lambda + 1, \lambda = \frac{x-1}{2}$$

$$\text{As } y = 3\lambda + 2, \lambda = \frac{y-2}{3}$$

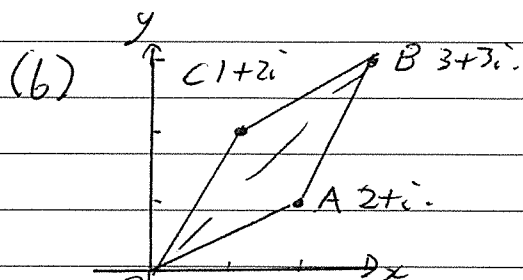
$$\text{As } z = 4\lambda - 1, \lambda = \frac{z+1}{4}$$

$$\therefore \frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{4} \quad (B)$$

Solutions: Y12 Trial: p.2

Q.(11.) (a) (i) $z = 1 - i\sqrt{3}$
 $= 2 \operatorname{cis}(-\frac{\pi}{3})$

(ii) $z^3 = 2^3 \operatorname{cis}(-\frac{3\pi}{3})$ (By De Moivre).
 $= 8 \operatorname{cis}(-\pi)$
 $= -8$, which is real.



(i) $\theta = 3 + 3i$

(ii) $\frac{\vec{OB}}{\vec{OA}} = \frac{3+3i}{2+i} \times \frac{(2-i)}{(2-i)}$
 $= \frac{9+3i}{5}$

$\operatorname{Arg}(\frac{\vec{OB}}{\vec{OA}}) = \tan^{-1} \angle AOB = \frac{(\frac{3}{5})}{(\frac{9}{5})}$
 $= \frac{1}{3}$

(iii) $\operatorname{Arg} \vec{OA} = \tan^{-1} \angle AOx = \frac{1}{2}$

$\therefore \operatorname{Arg}(\vec{OB}) = \tan^{-1} \angle BOx = \frac{1}{3} + \frac{1}{2}$

$\therefore \angle AOx + \angle AOB = \angle BOx$
 $\tan^{-1}(\frac{1}{2}) + \tan^{-1}(\frac{1}{3}) = \frac{\pi}{4}$

(c) (i) $(x+iy)^2 = 3-4i$, x, y real
 $(x^2-y^2) + 2ixy = 3-4i$

$\therefore x^2-y^2 = 3$ equating reals (1)

$2xy = -4$ equating imaginaries.

$y = -\frac{2}{x}$ (2)

(11)(c)(i) (continued):

Sub. (2) in (1):

$x^2 - \left(-\frac{2}{x}\right)^2 = 3$

$x^4 - 3x^2 - 4 = 0$

$(x^2-4)(x^2+1) = 0$

$x = \pm 2$, ($x \neq \pm i$ as x is real).

As $y = -\frac{2}{x}$, $y = \mp 1$.

$\therefore \sqrt{3-4i} = \pm(2-i)$

(ii) Solving $z^2 + (4-i)z + (3-i) = 0$

Noting $\Delta = b^2 - 4ac$

$= (4-i)^2 - 4 \times 1 \times (3-i)$

$= 3-4i$

$z = \frac{-(4-i) \pm \sqrt{\Delta}}{2 \times 1}$ [Quadratic formula]

As $\sqrt{\Delta} = \pm(2-i)$ [from (i)]

$z = \frac{-(4-i) \pm (2-i)}{2}$

$z = -1$ or $z = -3+i$

[Note that as the co-efficients in the original quadratic equation aren't real, complex solutions DON'T have to be conjugates of each other].

Solutions: Y12 Trial: p.3

$$Q.(12)(a)(i) \frac{-13x-10}{(x+1)^2(x-2)} = \frac{A}{(x+1)^2} + \frac{B}{(x+1)} + \frac{C}{(x-2)}$$

$$\therefore -13x-10 = A(x-2) + B(x+1)(x-2) + C(x+1)^2 \quad (1)$$

Sub. $x=2$ in (1):

$$-13 \times 2 - 10 = C(2+1)^2 \Rightarrow C = -4.$$

Sub. $x=-1$ in (1):

$$-13 \times -1 - 10 = A(-1-2) \Rightarrow A = -1.$$

Sub. $x=0$, $A=-1$, $C=-4$ in (1):

$$-10 = -2 \times -1 - 2B - 4.$$

$$4 = B$$

$$\therefore A = -1, B = 4 \text{ \& } C = -4 \text{ \& } \frac{-13x-10}{(x+1)^2(x-2)} = \frac{-1}{(x+1)^2} + \frac{4}{(x+1)} - \frac{4}{(x-2)}.$$

$$(ii) \int \frac{-13x-10}{(x+1)^2(x-2)} dx$$

$$= \int \left(\frac{-1}{(x+1)^2} + \frac{4}{(x+1)} - \frac{4}{(x-2)} \right) dx$$

$$= \frac{1}{x+1} + 4 \ln(x+1) - 4 \ln(x-2) + C.$$

$$(b) \int \frac{1}{\sin x - \cos x - 1} dx \quad \left[\begin{array}{l} \text{Letting } t = \tan\left(\frac{x}{2}\right) \\ \frac{dt}{dx} = \frac{1}{2} \sec^2\left(\frac{x}{2}\right) \\ = \frac{1+t^2}{2} \end{array} \right]$$

$$= \int \frac{1}{\frac{2t}{1+t^2} - \frac{(1-t^2)}{1+t^2} - 1} \cdot \frac{2}{1+t^2} dt \quad \left[\because \frac{dx}{dt} = \frac{dx}{dt} \cdot \frac{dt}{dt} = \frac{2}{1+t^2} \cdot dt \right]$$

$$= \int \frac{1}{2t-2} dt$$

$$= \frac{1}{2} \int \frac{2}{2t-2} dt$$

$$= \frac{1}{2} \int \frac{1}{t-1} dt$$

$$= \frac{1}{2} \ln(t-1) + C$$

Note also

$\frac{1}{2} \ln(2t-2) + C$ is also correct as it only differs from $\frac{1}{2} \ln(t-1)$ by $\frac{1}{2} \ln 2$ which is part of the constant C .

$$Q.(12)(c) \int e^x \cos x \cdot dx \quad \begin{array}{ll} u = e^x & v = \sin x \\ u' = e^x & v' = \cos x \end{array}$$

$$\text{Letting } I = \int e^x \cos x \cdot dx$$

$$I = e^x \sin x - \int e^x \sin x \cdot dx \quad (1)$$

$$\text{Taking } \int e^x \sin x \cdot dx \text{ out of (1): } \begin{array}{ll} u = e^x & v = -\cos x \\ u' = e^x & v' = \sin x \end{array}$$

$$= -e^x \cos x + \int e^x \cos x \cdot dx$$

$$\therefore \int e^x \sin x \cdot dx = -e^x \cos x + I \quad (2)$$

Sub. (2) in (1):

$$I = e^x \sin x - [-e^x \cos x + I]$$

$$2I = e^x [\sin x + \cos x]$$

$$\therefore I = \int e^x \cos x \cdot dx = \frac{1}{2} e^x [\sin x + \cos x] + C$$

$$(d)(i) \frac{d}{dx} (\cot x)$$

$$= \frac{d}{dx} \left(\frac{\cos x}{\sin x} \right)$$

$$= \frac{-\sin x \times \sin x - \cos x \times \cos x}{\sin^2 x} \quad (\text{By quotient rule})$$

$$= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x}$$

$$= -\frac{1}{\sin^2 x}$$

$$= -\operatorname{cosec}^2 x.$$

PTO $\rightarrow$

Q.(12)(d)(i) Let $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x \cdot dx$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^{n-2} x \cdot \cot^2 x \cdot dx$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^{n-2} x (\operatorname{cosec}^2 x - 1) \cdot dx$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^{n-2} x \cdot -\operatorname{cosec}^2 x \cdot dx - \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^{n-2} x \cdot dx$$

$$I_n = - \left[\frac{\cot^{n-1} x}{n-1} \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} - I_{n-2}$$

$$I_n = \left[\frac{0 - 1}{n-1} \right] - I_{n-2} \quad \left[\text{as } \cot \frac{\pi}{2} = 0, \cot \frac{\pi}{4} = 1 \therefore \cot^{n-1} \left(\frac{\pi}{2} \right) = 1^{n-1} = 1 \right]$$

$$= \frac{1}{n-1} - I_{n-2}$$

(iii) Finding I_6 .

$$I_0 = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^0 x \cdot dx$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 1 \cdot dx$$

$$= \frac{\pi}{4}$$

$$I_2 = 1 - I_0$$

$$= 1 - \frac{\pi}{4}$$

$$I_4 = \frac{1}{3} - I_2$$

$$= \frac{1}{3} - \left(1 - \frac{\pi}{4} \right)$$

$$= \frac{1}{3} - 1 + \frac{\pi}{4}$$

$$= -\frac{2}{3} + \frac{\pi}{4}$$

$$I_6 = \frac{1}{5} - I_4$$

$$= \frac{1}{5} + \frac{2}{3} - \frac{\pi}{4}$$

$$= \frac{13}{15} - \frac{\pi}{4}$$

(iv) Using $I_4 = \frac{1}{3} - 1 + \frac{\pi}{4}$

$$I_6 = \frac{1}{5} - \frac{1}{3} + 1 - \frac{\pi}{4}$$

$$I_8 = \frac{1}{7} - \frac{1}{5} + \frac{1}{3} - 1 + \frac{\pi}{4}$$

$$I_{10} = \frac{1}{9} - \frac{1}{7} + \frac{1}{5} - \frac{1}{3} + 1 - \frac{\pi}{4}$$

$$I_{2n+2} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots + \frac{1}{2n+1} - \frac{\pi}{4}$$

As $I_n \rightarrow 0$ as $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \left(1 - \frac{1}{3} + \frac{1}{5} - \dots + \frac{1}{2n+1} - \frac{\pi}{4} \right) = 0$$

$$\therefore \lim_{n \rightarrow \infty} \left(1 - \frac{1}{3} + \frac{1}{5} - \dots + \frac{1}{2n+1} \right) = \frac{\pi}{4}$$

Q. (13)(a) Contrapositive of "if $n^2 + 6n$ is even, n is even" is
 "If n is odd, $n^2 + 6n$ is odd"

Letting n be odd i.e. $n = 2k-1$, k an integer.

$$\therefore n^2 + 6n$$

$$= (2k-1)^2 + 6(2k-1)$$

$$= 4k^2 + 8k - 5$$

$$= 2[2k^2 + 4k - 2] - 1$$

which is odd as $2(2k^2 + 4k - 2)$ is even.

(b) Let $\log_3 11$ be rational

i.e. $\log_3 11 = \frac{p}{q}$, p, q integers.

$$3^{\frac{p}{q}} = 11$$

$$3^p = 11^q$$

which is not possible as both 3 & 11 are prime

& any whole number can only have 1 set of prime factors [fundamental theorem of arithmetic]

$\therefore \log_3 11$ is irrational.

(c) Step 1: Show true for $n=1$.

$$\begin{array}{ll} \text{LHS:} & \text{RHS:} \\ = 3^1 & = 1^2 \end{array}$$

$$= 3 = 1$$

$$\text{LHS} > \text{RHS}$$

True for $n=1$.

Step 2: Assume true for $n=k$

i.e. $3^k > k^2$, k an integer ≥ 1 .

Step 3: Prove true for $n=k+1$

i.e. $3^{k+1} > (k+1)^2$, k an integer ≥ 1 .

10:56

LHS:

$$3^{k+1} = 3 \times 3^k$$

$$> 3k^2 \text{ [by assumption]}$$

$$> (k+1)^2 \text{ [by (2) below]}$$

$\therefore$ If it is true for $n=k$ it will be true for $n=k+1$

$\therefore$ As it is true for $n=2$ it will be true for $n=2+1=3$ & so on for all positive integers k . It was also shown to be true for $n=1$ in Step 1

$\therefore 3 > n^2$ for all positive integers n .

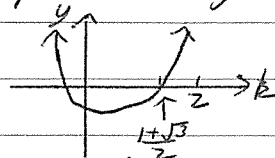
Showing $3k^2 > (k+1)^2$

$$\text{Solve } 3k^2 - (k+1)^2 > 0$$

$$2k^2 - 2k - 1 > 0$$

$$\text{For } 2k^2 - 2k - 1 = 0, k = \frac{1 \pm \sqrt{3}}{2}$$

From diagram, $3k^2 > (k+1)^2, k \geq 2$.



Q. (13)(d) Let $10x + y = 17k$, k an integer
 $y = 17k - 10x$. (1)

$$\begin{aligned} & \therefore 3y - 4x \\ &= 3(17k - 10x) - 4x \\ &= 51k - 30x - 4x \\ &= 51k - 34x \\ &= 17(3 - 2x) \end{aligned}$$

$= 17L$, $L = 3 - 2x$, an integer as x is an integer.

$\therefore 3y - 4x$ is also divisible by 17.

(e)(i) If a, b real,

$$\begin{aligned} (a^2 - b^2)^2 &\geq 0 \\ a^4 + b^4 - 2a^2b^2 &\geq 0 \\ a^4 + b^4 &\geq 2a^2b^2 \quad (1) \end{aligned}$$

(ii) Using (i) above,

$$\begin{aligned} a^4 + b^4 &\geq 2a^2b^2 \quad \& \quad c^4 + d^4 \geq 2c^2d^2 \\ \therefore a^4 + b^4 + c^4 + d^4 &\geq 2(a^2b^2 + c^2d^2) \\ &\geq 2(2abcd) \quad [a^2b^2 + c^2d^2 \geq 2abcd \text{ by (i)}] \\ &= 4abcd. \end{aligned}$$

(iii) Letting $w = a^4$, $x = b^4$, $y = c^4$ & $z = d^4$,

$$w + x + y + z \geq 4 \sqrt[4]{w} \times \sqrt[4]{x} \times \sqrt[4]{y} \times \sqrt[4]{z} \quad \div 4$$

$$\frac{w + x + y + z}{4} \geq \sqrt[4]{wxyz}.$$

Q.14)(a)(i) $\underline{r} \cdot \underline{q}$
 $= (\underline{i} + 2\underline{j} - \underline{k}) \cdot (2\underline{i} - \underline{j} + \underline{k})$
 $= 2 - 2 - 1$
 $= -1.$

(ii) $\cos \angle POQ = \frac{\underline{r} \cdot \underline{q}}{|\underline{r}| |\underline{q}|}$
 $= \frac{-1}{\sqrt{6} \times \sqrt{6}}$
 $= \frac{-1}{6}.$

$\angle POQ = 99^{\circ}36'$ [nearest minute].

(iii) $\text{Proj}_{\underline{r}} \underline{q} = \frac{\underline{r} \cdot \underline{q}}{|\underline{r}|^2} \underline{r}$
 $= -\frac{1}{6} (\underline{i} + 2\underline{j} - \underline{k}).$

(b)(i) $(12-10)^2 + (-6+12)^2 + (6-14)^2 = 104.$

(ii) The line $\begin{pmatrix} 12 \\ -6 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix}$ forms a diameter if it passes through the centre of the sphere (the point $\begin{pmatrix} 10 \\ -12 \\ 14 \end{pmatrix}$).

Showing this, $\begin{pmatrix} 12 \\ -6 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 10 \\ -12 \\ 14 \end{pmatrix}$

$12 + \lambda = 10 \Rightarrow \lambda = -2$

$-6 + 3\lambda = -12 \Rightarrow \lambda = -2$

$6 - 4\lambda = 14 \Rightarrow \lambda = -2.$

Line passes through centre. By symmetry, other point of intersection will be $\begin{pmatrix} 8 \\ -18 \\ 22 \end{pmatrix}.$

(iii) Checking to see if lines are skew:

$\begin{pmatrix} 12 \\ -6 \\ 6 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} = \begin{pmatrix} -8 \\ 12 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 6 \\ -8 \\ 5 \end{pmatrix}$

$12 + \lambda_1 = -8 + 6\lambda_2 \Rightarrow \lambda_1 - 6\lambda_2 = -20 \quad (1)$
 $-6 + 3\lambda_1 = 12 - 8\lambda_2 \Rightarrow 3\lambda_1 + 8\lambda_2 = 18 \quad (2)$
 $6 - 4\lambda_1 = 2 + 5\lambda_2 \Rightarrow -4\lambda_1 - 5\lambda_2 = -4$
 $0 - 4\lambda_1 + 5\lambda_2 = 4 \quad (3)$

Using (1) & (2)
 $(1) \times 3 = (3)$
 $3\lambda_1 + 8\lambda_2 = 18 \quad (2)$
 $3\lambda_1 - 18\lambda_2 = -60 \quad (3)$

$$26\lambda_2 = 78$$

$$\lambda_2 = 3$$

Sub. $\lambda_2 = 3$ in (1)

$$\lambda_1 - 18 = -20$$

$$\lambda_1 = -2$$

Sub. $\lambda_1 = -2, \lambda_2 = 3$ in (3)

to see if it works:

$$4 \times -2 + 5 \times 3 = 7 \neq 4$$

→ Lines don't intersect.

If you used (1) & (3)
 $(1) \times 4 = (5)$

$$4\lambda_1 + 5\lambda_2 = 4 \quad (3)$$

$$4\lambda_1 - 24\lambda_2 = -80 \quad (5)$$

$$29\lambda_2 = 84$$

$$\lambda_2 = \frac{84}{29} = 2\frac{26}{29}$$

Sub. $\lambda_2 = 2\frac{26}{29}$ in (1)

$$\lambda_1 - 6 \times 2\frac{26}{29} = -20$$

$$\lambda_1 = -2\frac{18}{29}$$

Sub. $\lambda_1 = -2\frac{18}{29}, \lambda_2 = 2\frac{26}{29}$

in (2):
 $3 \times -2\frac{18}{29} + 8 \times 2\frac{26}{29} = 15\frac{9}{29} \neq 18$

→ Lines don't intersect.

If you used (2) & (3)

$$(2) \times 4 = (6) \quad (3) \times 3 = (7)$$

$$12\lambda_1 + 32\lambda_2 = 72 \quad (6)$$

$$12\lambda_1 + 15\lambda_2 = 12 \quad (7)$$

$$17\lambda_2 = 60$$

$$\lambda_2 = \frac{60}{17} = 3\frac{9}{17}$$

Sub. $\lambda_2 = 3\frac{9}{17}$ in (3)

$$4\lambda_1 + 5 \times 3\frac{9}{17} = 4$$

$$\lambda_1 = -\frac{53}{17} = -3\frac{7}{17}$$

Sub. $\lambda_1 = -3\frac{7}{17}, \lambda_2 = 3\frac{9}{17}$ in (1)

to see if it works:

$$-3\frac{7}{17} - 6 \times 3\frac{9}{17} \neq$$

$$-24\frac{10}{17} \neq -20$$

→ Lines don't intersect.

By observation, as $\begin{pmatrix} 6 \\ -8 \\ 5 \end{pmatrix} \neq \lambda \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix}$, direction vectors aren't //.

→ Lines don't intersect & aren't // . They are skew.

$$(iv) (-8 + 6\lambda - 10)^2 + (12 - 8\lambda + 12)^2 + (2 + 5\lambda - 14)^2 = 104$$

$$36(\lambda - 3)^2 + 64(3 - \lambda)^2 + (5\lambda - 12)^2 = 104$$

$$100(\lambda - 3)^2 + (5\lambda - 12)^2 = 104$$

$$125\lambda^2 - 720\lambda + 1044 = 104$$

$$25\lambda^2 - 144\lambda + 188 = 0$$

$$\lambda = \frac{144 \pm \sqrt{(-144)^2 - 4 \times 25 \times 188}}{2 \times 25}$$

$$\lambda = \frac{94}{25} \text{ or } \lambda = 2$$

Points of intersection of line & sphere

$$= \begin{pmatrix} -8 \\ 12 \\ 2 \end{pmatrix} + \frac{94}{25} \begin{pmatrix} 6 \\ -8 \\ 5 \end{pmatrix} = \begin{pmatrix} 14\frac{14}{25} \\ -18\frac{2}{25} \\ 4\frac{4}{25} \end{pmatrix} \text{ or } = \begin{pmatrix} -8 \\ 12 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} 6 \\ -8 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ -4 \\ 12 \end{pmatrix}$$

$$(v) \begin{pmatrix} -8 \\ 12 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -8 \\ 5 \end{pmatrix} \text{ passes through } \begin{pmatrix} 10 \\ -12 \\ z \end{pmatrix} \quad 6\lambda - 8 = 10 \Rightarrow \lambda = 3$$

$$\text{Check: } -8\lambda + 12 = -12 \Rightarrow \lambda = 3$$

So if $\lambda = 3$, z co-ordinate = $2 + 5 \times 3 = 17$.

Point above centre of sphere line passes through is $\begin{pmatrix} 10 \\ -12 \\ 17 \end{pmatrix}$.

12 Ext 2 Paper p.10

Q. (15) (a) → Starts at BOTTOM so $-\cos$

(i) → Period = $\frac{2\pi}{n} = \frac{13}{n}$ hours.

$$\therefore \frac{2\pi}{n} = \frac{13}{n} = 13n$$

$$n = \frac{2\pi}{13}$$

→ Centre of "motion" = 9. Amplitude = 4.

∴ Depth is $x = -4\cos\left(\frac{2\pi t}{13}\right) + 9$.

(ii) Finding when $-4\cos\left(\frac{2\pi t}{13}\right) + 9 = 8$.

$$\cos\left(\frac{2\pi t}{13}\right) = \frac{1}{4}$$

$$\frac{2\pi t}{13} = 1.1071 \dots \text{ or } 4.9650 \dots$$

$$t = 2\text{h } 43\text{mins } 37\text{s. or } 10\text{h } 16\text{mins } 22\text{s.}$$

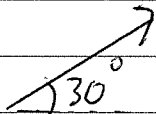
But to the nearest minute must be BETWEEN these times.

So $t = 9:44\text{a.m. to } 5:16\text{p.m.}$

(b)

(i) Initial horizontal velocity:

$$\text{Force} = 5\ddot{x} = -\frac{5}{6}v = ma$$



$$\ddot{x} = -\frac{1}{6}v$$

$$F = ma = 5\ddot{y} = -50 - \frac{5}{6}v$$

$$\ddot{y} = -10 - \frac{1}{6}v$$

(ii) Initial $\ddot{x}$ & $\ddot{y}$:

$$\ddot{y} = 600 \sin 30^\circ = 300 \text{ m/s}^2$$

$$\ddot{x} = 600 \cos 30^\circ = 300\sqrt{3} \text{ m/s}^2$$

Y12 Ext 2 p.11

Q.(15)(b)(iii) $\ddot{x} = \frac{dv}{dt} (v = v_x) = -\frac{1}{6}v$

$$\therefore \frac{dt}{dv} = -\frac{6}{v}$$

$$t = \int \frac{-6}{v} dv$$

$$= -6 \ln v + C$$

As $v = 300\sqrt{3}$ when $t = 0$,

$$0 = -6 \ln 300\sqrt{3} + C$$

$$t = -6 \ln v + 6 \ln (300\sqrt{3})$$

$$= -6 \ln \left(\frac{v}{300\sqrt{3}} \right)$$

$$\therefore e^{-\frac{t}{6}} = \frac{v}{300\sqrt{3}}$$

$$300\sqrt{3} e^{-\frac{t}{6}} = v$$

$$x = \int 300\sqrt{3} e^{-\frac{t}{6}} dt$$

$$= -1800\sqrt{3} e^{-\frac{t}{6}} + C$$

As $x = 0$ when $t = 0$

$$0 = -1800\sqrt{3} e^0 + C$$

$$x = 1800\sqrt{3} (1 - e^{-\frac{t}{6}})$$

(iv) As $t \rightarrow \infty$, $e^{-\frac{t}{6}} \rightarrow 0$ & $x \rightarrow 1800\sqrt{3}$.

Limiting range = $1800\sqrt{3}$ m.

(v) $\ddot{y} = \frac{dv_y}{dt} = -10 - \frac{1}{6}v$

$$\frac{dt}{dv} = -\frac{(60+v)}{6}$$

$$\frac{dt}{dv} = -\frac{6}{60+v}$$

$$t = \int \frac{-6}{60+v} dv$$

$$= -6 \ln(60+v) + C$$

As $y = 300$ when $t = 0$

$$0 = -6 \ln(60+300) + C$$

$$6 \ln 360 = C$$

$$t = -6 \ln(60+v) + 6 \ln 360$$

$$t = -6 \ln \left(\frac{60+v}{360} \right)$$

$$e^{-\frac{t}{6}} = \frac{60+v}{360}$$

$$360 e^{-\frac{t}{6}} - 60 = v$$

Y12 Ext. 2 p.12.

Q.(15)(b)(v) (continued):

$$y = \int 360e^{-\frac{t}{6}} - 60 \cdot dt$$

$$= -2160e^{-\frac{t}{6}} - 60t + C$$

As $y = 0$ when $t = 0$

$$0 = -2160e^0 - 60 \times 0 + C$$

$$2160 = C$$

$$y = -2160e^{-\frac{t}{6}} - 60t + 2160.$$

(vi) Maximum height: Is y when $y' = 0$

$$360e^{-\frac{t}{6}} - 60 = 0$$

$$e^{-\frac{t}{6}} = \frac{1}{6}.$$

$$\& -\frac{t}{6} = \ln\left(\frac{1}{6}\right)$$

$$t = -6\ln\left(\frac{1}{6}\right)$$

$$= 6\ln 6.$$

$$\text{Max. height} = -2160e^{-\ln 6} - 60 \times 6\ln 6 + 2160$$

$$= 1154.966 \dots \text{m.}$$

So the maximum height the projectile reaches is

approx. 1155m after 10.75 seconds.

Y12 Ext 2 p.13

Q.16)(a)(i)

$$\ddot{x} = v \cdot \frac{dv}{dx} = -g - kv^2$$

$$\frac{dv}{dx} = \frac{-g - kv^2}{v}$$

$$\frac{dx}{dv} = \frac{-v}{g + kv^2}$$

$$x = -\frac{1}{2k} \int \frac{2kv}{g + kv^2} \cdot dv$$

$$x = -\frac{1}{2k} \ln(g + kv^2) + C$$

As $x = 0$ when $v = U$,

$$0 = -\frac{1}{2k} \ln(g + kU^2) + C$$

$$\frac{1}{2k} \ln(g + kU^2) = C$$

$$x = -\frac{1}{2k} \ln(g + kv^2) + \frac{1}{2k} \ln(g + kU^2)$$

$$x = \frac{1}{2k} \ln\left(\frac{g + kU^2}{g + kv^2}\right)$$

(ii) Maximum height reached $v = 0$

$$x = \frac{1}{2k} \ln\left(\frac{g + kU^2}{g}\right)$$

$$x = \frac{1}{2k} \ln\left(1 + \frac{kU^2}{g}\right)$$

(iii)

$$\begin{array}{c} \downarrow \\ g \\ kv^2 \\ \uparrow \end{array}$$

Going down: $\ddot{x} = g - kv^2$

$$v \cdot \frac{dv}{dx} = g - kv^2$$

$$\frac{dv}{dx} = \frac{g - kv^2}{v}$$

$$\frac{dx}{dv} = \frac{v}{g - kv^2}$$

$$x = -\frac{1}{2k} \int \frac{2kv}{g - kv^2} \cdot dv$$

$$x = -\frac{1}{2k} \ln(g - kv^2) + C$$

As projectile is falling from rest, $0 = -\frac{1}{2k} \ln g + C \Rightarrow C = \frac{1}{2k} \ln g$

$$x = \frac{1}{2k} \ln\left(\frac{g}{g - kv^2}\right)$$

Q. (16)(a)(iv) T = terminal velocity [when $\ddot{x} = 0$].

$$g - kv^2 = 0$$

$$v^2 = \frac{g}{k}$$

$$v = \sqrt{\frac{g}{k}}$$

$$\therefore T = \sqrt{\frac{g}{k}}$$

W is impact velocity $\rightarrow$ velocity when projectile hits the ground, which is where $x = \frac{1}{2k} \ln\left(1 + \frac{kU^2}{g}\right)$

$$\text{So } \frac{1}{2k} \ln\left(\frac{g}{g - kW^2}\right) = \frac{1}{2k} \ln\left(1 + \frac{kU^2}{g}\right)$$

$$\frac{g}{(g - kW^2)} = 1 + \frac{kU^2}{g}$$

$$\begin{aligned} & \times g(g - kW^2) \\ g^2 &= g^2 - gkW^2 + kU^2g - k^2U^2W^2 \\ & + gkW^2 - g^2 + k^2U^2W^2 \end{aligned}$$

$$gkW^2 + k^2U^2W^2 = kU^2g$$

$$\frac{1}{U^2} + \frac{k}{g} \stackrel{\div \text{BS by } gkW^2U^2}{=} \frac{1}{W^2}$$

$$\text{As } T (\text{Terminal Velocity}) = \sqrt{\frac{g}{k}}, \quad \frac{1}{T^2} = \frac{k}{g}$$

$$\therefore \frac{1}{U^2} + \frac{1}{T^2} = \frac{1}{W^2} \quad \text{QED.}$$

$$(b)(i) z^5 - 1 = 0$$

$$z = \text{cis } \frac{2\pi}{5}, \text{cis } \frac{4\pi}{5}, \text{cis } \frac{6\pi}{5}, \text{cis } \frac{8\pi}{5}, 1.$$

(ii) w is $\text{cis } \frac{2\pi}{5}$ [root with smallest positive argument].

Other roots = $w^2 (\text{cis } \frac{4\pi}{5})$, $w^3 (\text{cis } \frac{6\pi}{5})$, $w^4 (\text{cis } \frac{8\pi}{5})$ & 1.

$\therefore$ By sum of roots of $z^5 - 1 = 0$, $1 + w + w^2 + w^3 + w^4 = 0$

$$w + w^2 + w^3 + w^4 = -1.$$

$$\text{As } \frac{w^4}{w^5} = \frac{w^4}{1} = \frac{1}{w} \text{ \& } \frac{w^3}{w^5} = \frac{w^3}{1} = w^2, w + w^4 + w^2 + w^3 = -1.$$

$$\left(w + \frac{1}{w}\right) + \left(w^2 + \frac{1}{w^2}\right) = -1.$$

Y12 Ext. 2 p.15

$$\begin{aligned} \text{Q.(16)(b)(iii) As } w + \frac{1}{w} &= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \left(-\frac{2\pi}{5} \right) + i \sin \left(-\frac{2\pi}{5} \right) \\ w &= \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5} \\ &= 2 \cos \frac{2\pi}{5} \quad (\text{as cos even, sin odd}). \end{aligned}$$

$$\text{\& similarly } w^2 + \frac{1}{w^2} = 2 \cos \frac{4\pi}{5}.$$

$$2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5} = -1.$$

$$\text{Given that } \cos \frac{4\pi}{5} = 2 \cos^2 \left(\frac{2\pi}{5} \right) - 1 \quad [\text{by } \cos 2\theta = 2 \cos^2 \theta - 1],$$

$$2 \cos \frac{2\pi}{5} + 4 \cos^2 \frac{2\pi}{5} - 2 = -1.$$

$$4 \cos^2 \frac{2\pi}{5} + 2 \cos \frac{2\pi}{5} - 1 = 0.$$

$$\text{Hence } \cos \frac{2\pi}{5} \text{ is a root of } 4x^2 + 2x - 1 = 0$$

$$\text{(iv) Solving } 4x^2 + 2x - 1 = 0$$

$$\begin{aligned} x &= \frac{-2 \pm \sqrt{2^2 - 4 \times 4 \times -1}}{2 \times 4} \\ &= \frac{-2 \pm \sqrt{20}}{8} \\ &= \frac{-2 \pm 2\sqrt{5}}{8} \\ &= \frac{-1 \pm \sqrt{5}}{4}. \end{aligned}$$

As $\frac{2\pi}{5}$ is in Q_1 , $\cos \frac{2\pi}{5}$ is positive.

$$\therefore \cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4}.$$

END OF SOLUTIONS!!!